Community energy: close the gap with local operations

Brief for Colombian utilities and foundations: public energy-poverty data, the real cost of rural OPEX, and cooperatives as a sustainability vehicle.

1. The national gap

15.4%
IMPE incidence 2024 (~8.1 M people in energy poverty)[1]

9.5%
2030 scenario target (~5.17 M people; −2.9 M vs 2024)[1]

75.92% / 93.12%
Rural vs national residential electric coverage (ICEE 2024)[2]

63.1% / 52.8%
Departmental IMPE La Guajira / Chocó (persons, 2024)[1]

At the observed pace (~300,000 people per year leaving energy poverty), lifting ~2.9 M people takes ~9.7 years from 2024 (single-digit incidence ~2034).[1] The IMPE 2030 scenario needs ~490,000 people per year × 6 years (≥1.0 pp/year incidence).[1]

Pace ~300k people/year
2.9 M ÷ 300k ≈ 9.7 years
~2034

Scenario ~490k people/year
490k × 6 ≈ 2.94 M
2030

2. Reframe: OPEX, not CAPEX alone

- **Contraloría:** sustainability concern where methodologies to pay for operation, microgrid administration, and off-grid PV are missing; maintenance and monitoring failures.[3]
- **Amazonas:** ~COP 30 billion (royalties) solar investment not operating; damaged batteries — "critical state" on sustainability.[4]
- **CREG:** thousands of rural PV systems without management → "often abandoned"; missing standards, certified installers, and O&M.[5]

[1] Multidimensional Energy Poverty Index (IMPE), 2025 report, 3rd edition (2024 base year), pp. 12, 22, 33.

[2] UPME, ICEE Bulletin 2024; UPME press release on 93.12% coverage (2024).

[3] El Colombiano, Contraloría on solar projects in remote areas.

[4] Seguimiento.co / Procuraduría, abandoned Amazonas project.

[5] CREG GEN-Renewables report (Circular 005-13); CREG Concept 4022/2019 (O&M).

3. The cooperative as the tool

CTH PROPOSAL

O&M services cooperative and green jobs complementing CAPEX (FENOGE, IPSE, PIEC) with:

- Explicit OPEX contracts (PV cleaning, batteries, remote monitoring) aligned with CREG/SSPD rules.
- Local revenue and ownership: community governance in energy communities so assets are not orphaned after commissioning.
- Tracking with IMPE indicators (access and quality) — does not replace official ICEE or IPEM.

4. Local talent and green jobs

Proposed collaboration with Escuela Colombiana de Ingeniería Julio Garavito — *no published framework agreement* with CleantechHUB.

- Documented academic engagement: CLSA chair (energy-sector challenges); 3rd place Solucionatón 10X Connect Bogotá (Oct 2026).[6]
- Training offer: renewable-energy diploma and courses, energy-transition and circular-PV modules (incl. O&M); 101.76 kWp lab and off-grid ZNI bench.[7]

5. Territorial priority (IMPE)

Departments with high IMPE incidence (persons, 2024) guide pilot corridors: La Guajira **63.1%**, Chocó **52.8%**; La Guajira ICEE **54.28%**.[1,2]

Productive territorial partner lists and named pilot sites are omitted pending additional sourced evidence in the evidence pack.

7. Dutch partner network (target)

CleantechHUB maps **finance and peer-learning channels** — *no signed agreements* with Energy Coop:

- **FMO:** solar project finance in Colombia (e.g. 252 MWp utility-scale Cesar/Córdoba; ~52.8 MWp mini-farm portfolio).[8]
- **RVO** (Netherlands Enterprise Agency): commercial facilitation in the clean-energy ecosystem (La Guajira solar MOU context).[9]

[6] escuelaing.edu.co — CLSA and Solucionatón 10X news (2026).

[7] escuelaing.edu.co — continuing-education programs and Renewable Energy Laboratory.

[8] fmo.nl — Colombia project news and sheets (La Mata / La Unión; Sol de la Sierra / Ayura).

[9] Gutami MOU 2023; Holland House Colombia (facilitation).

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